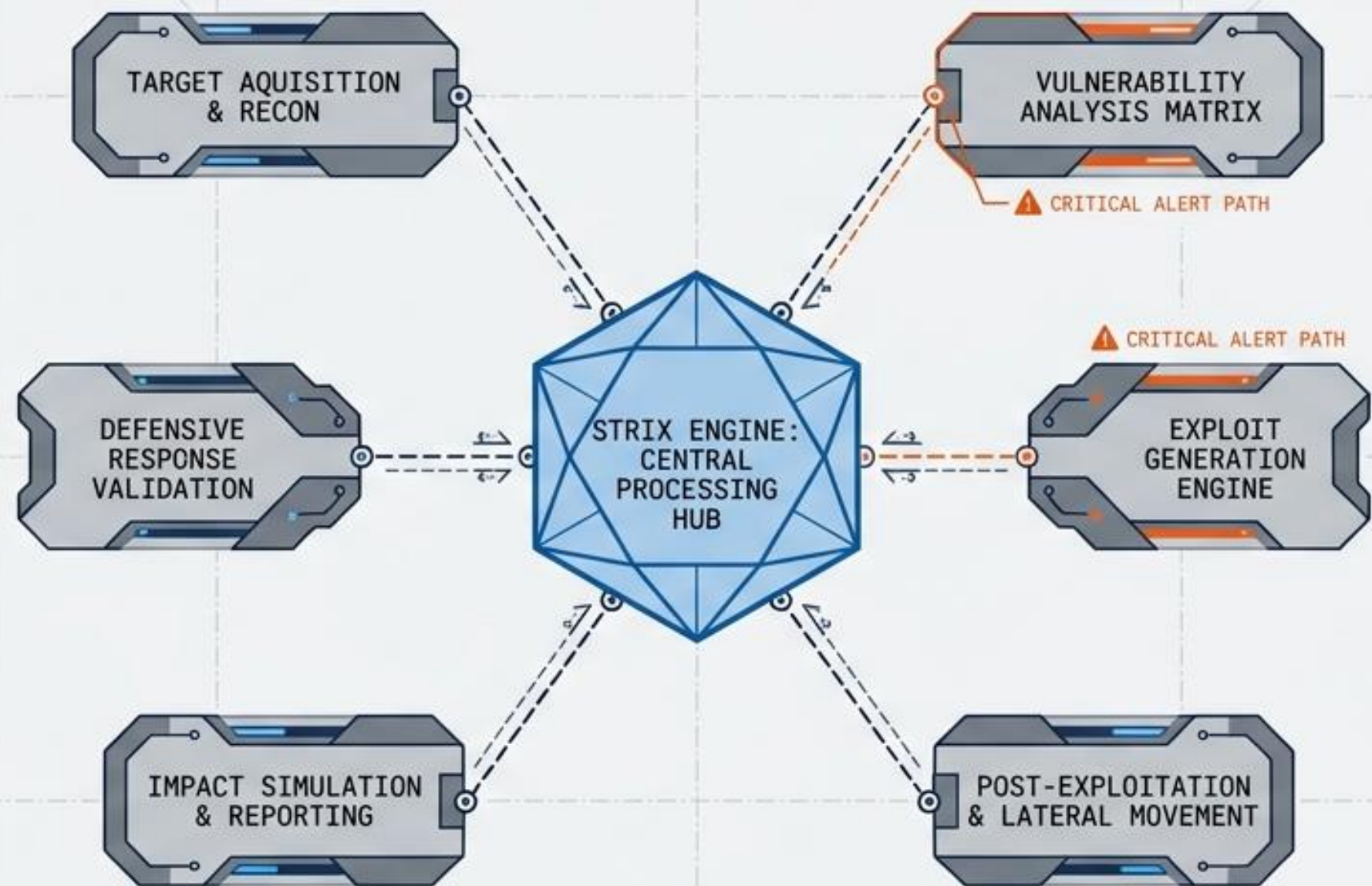


Architecting the Autonomous AI Red Team

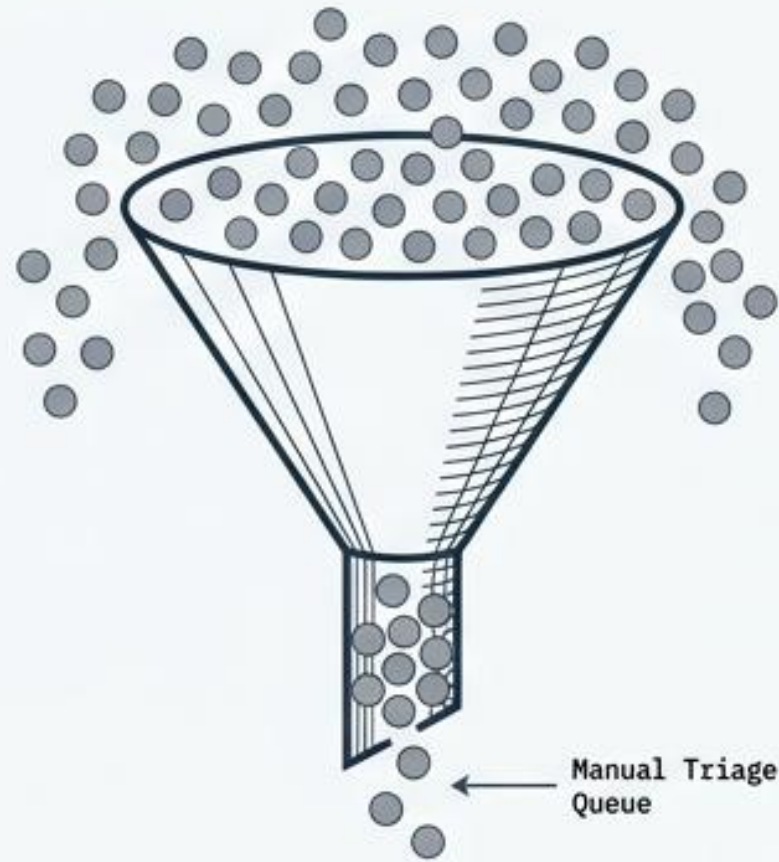
Deep Dive into the Strix Engine: Scalability, Extensibility, and Deployment

Leadership Briefing



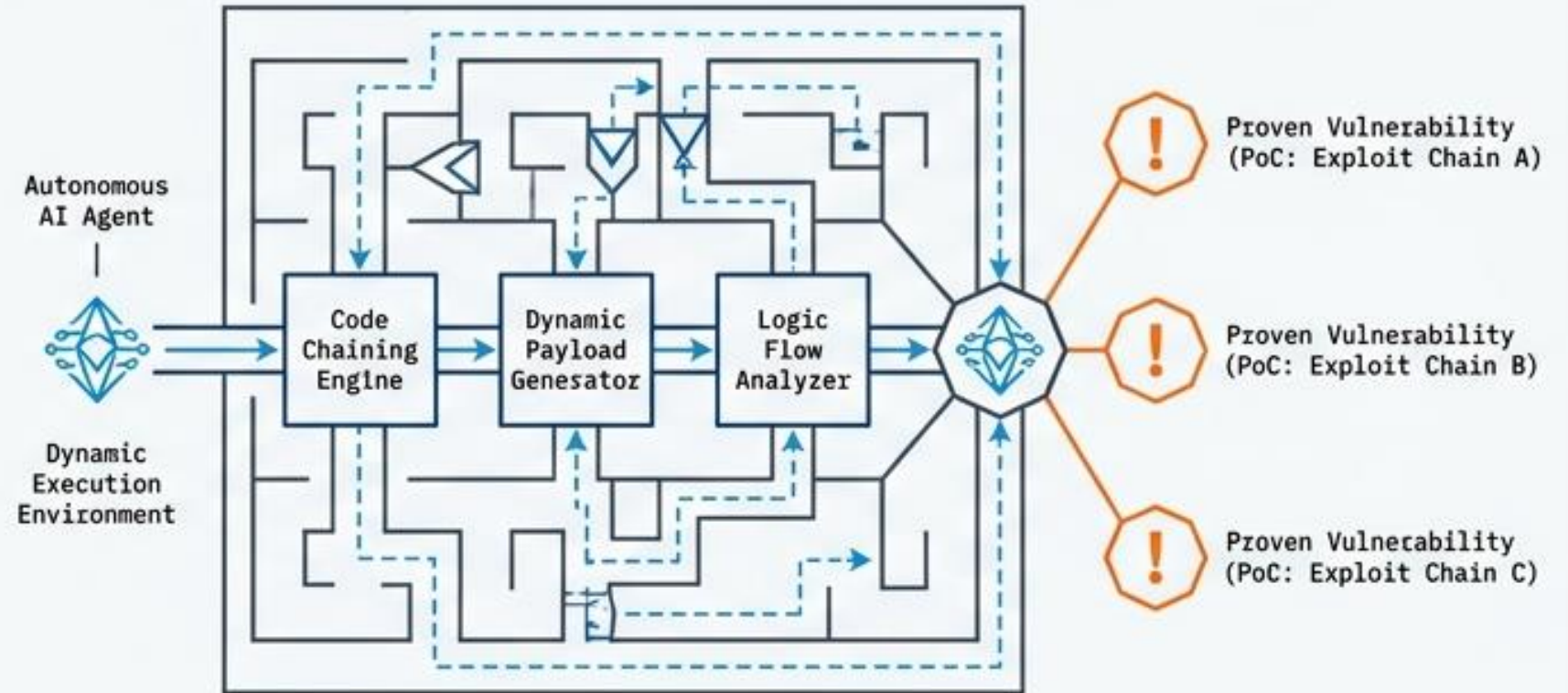
Moving Beyond Noise and Bottlenecks

Legacy Scanners



Legacy SAST/DAST tools create unscalable noise. Manual pentesting is too slow for modern CI/CD.

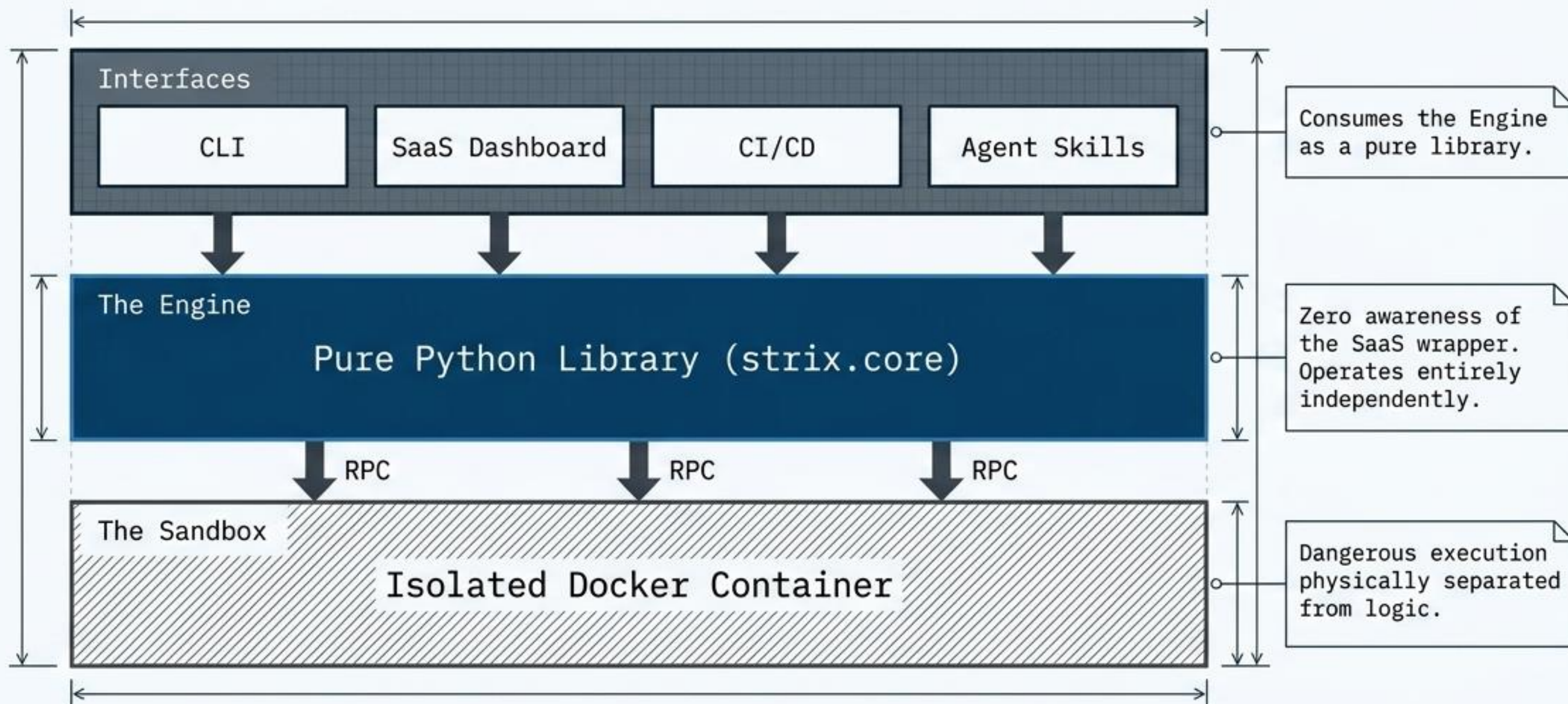
The Strix Paradigm



Autonomous AI agents act like real hackers dynamically running code, chaining exploits, and proving vulnerabilities with working Proofs-of-Concept.

The Architectural Imperative: Build an engine unbounded in scale, completely isolated in execution, and universally deployable.

The Isolation Boundary



Why It Matters: The core engine remains open and reusable. Enterprise features scale independently in the SaaS layer without creating merge conflicts.

Unbounded Scale and Economic Safety

Unbounded Concurrency

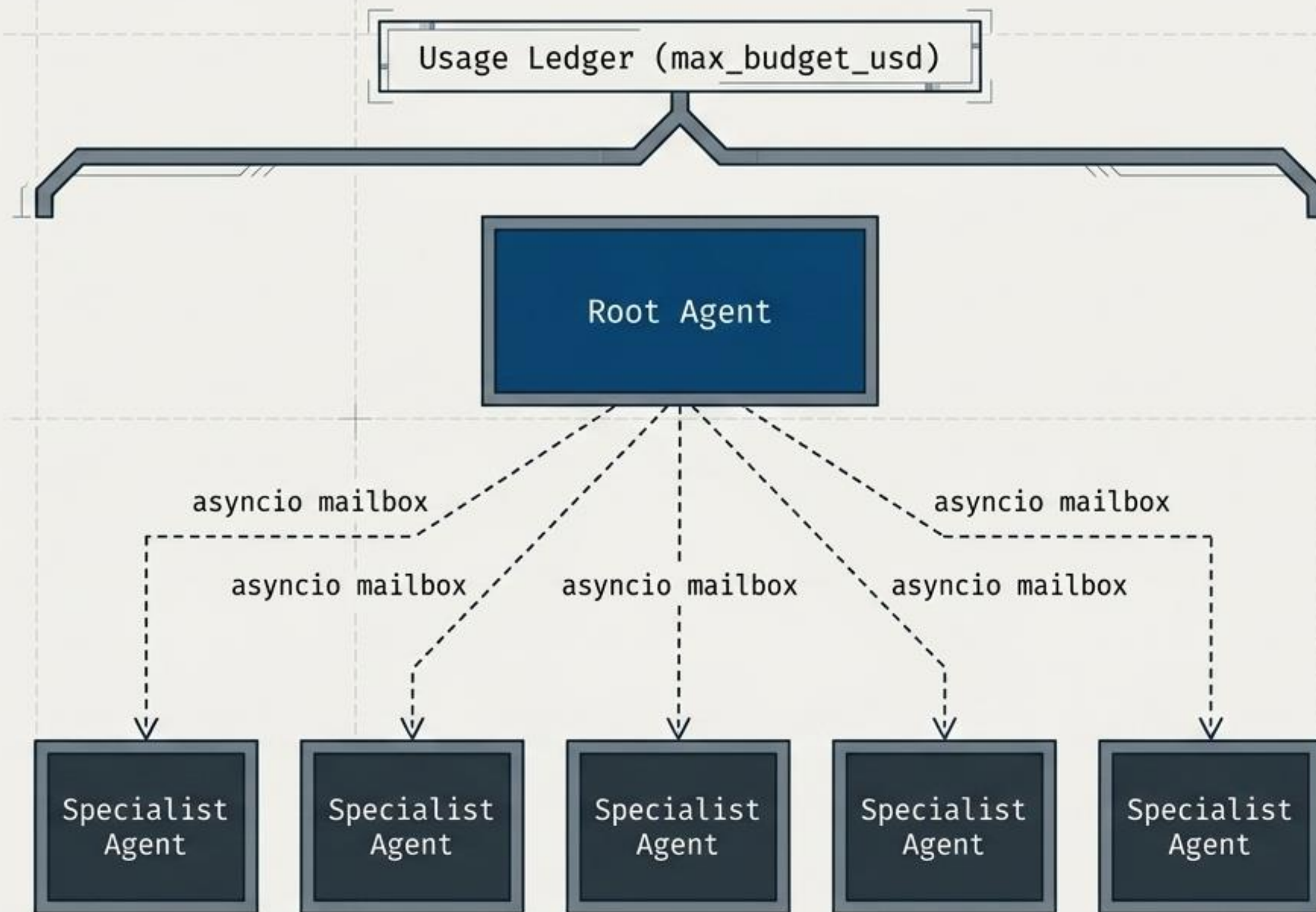
Every agent is a separate `asyncio.Task` on a single event loop. Coordination occurs dynamically via mailbox queues, not blocking calls.

Shared Sandbox State

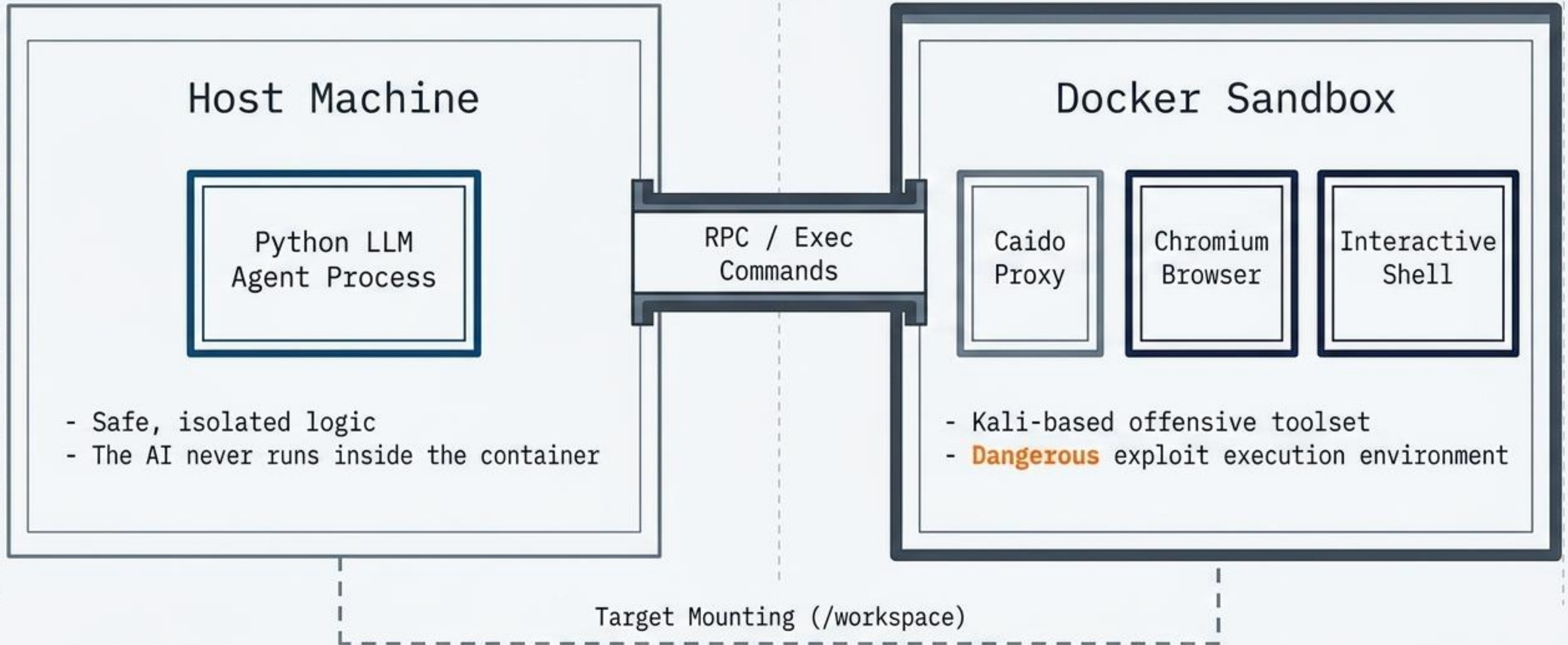
The entire agent tree shares a single Docker sandbox session, ensuring all parallel agents operate on the same state.

Economic Circuit Breaker

If the scan hits 90% of the shared budget, child agents are immediately halted. The final 10% is strictly reserved for the Root Agent to finalize the report.

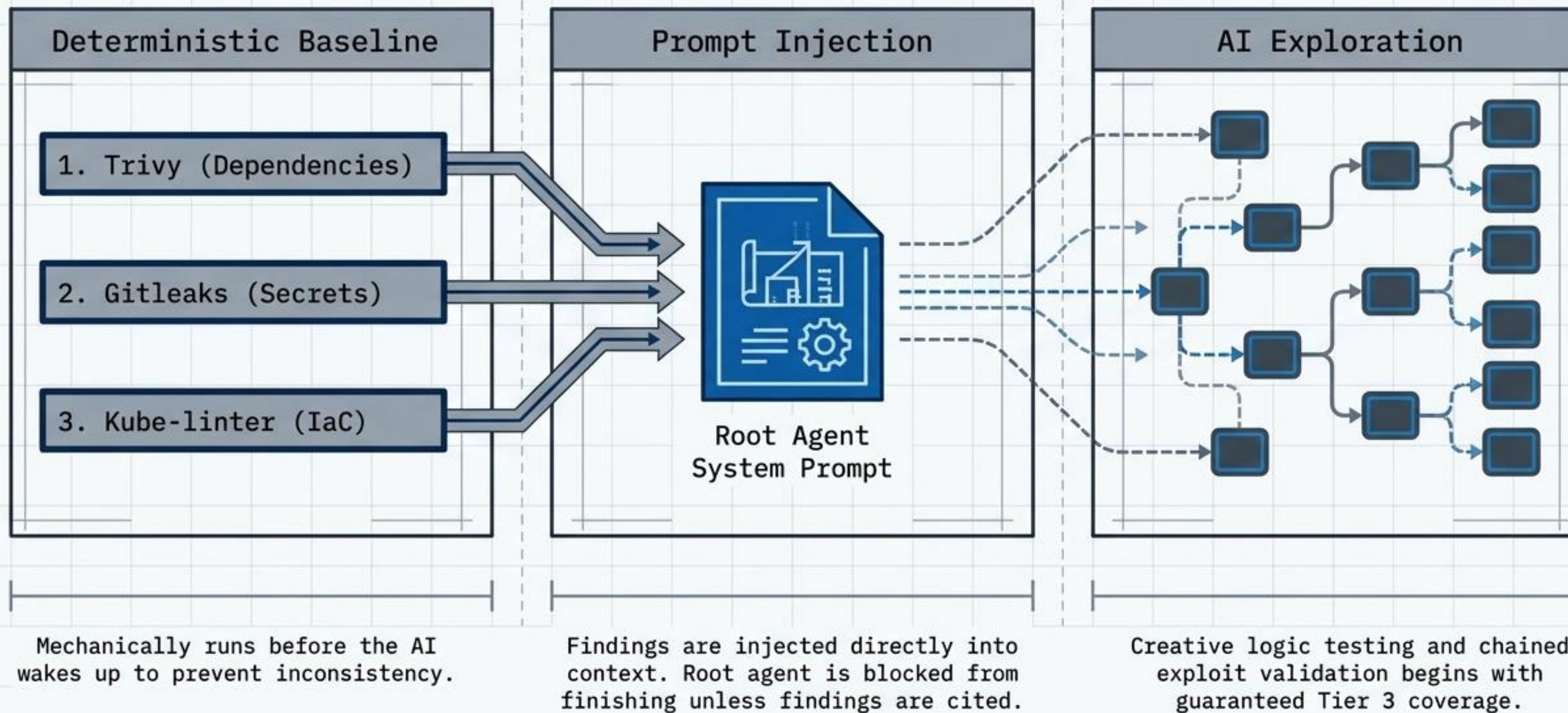


The Host-Sandbox Split

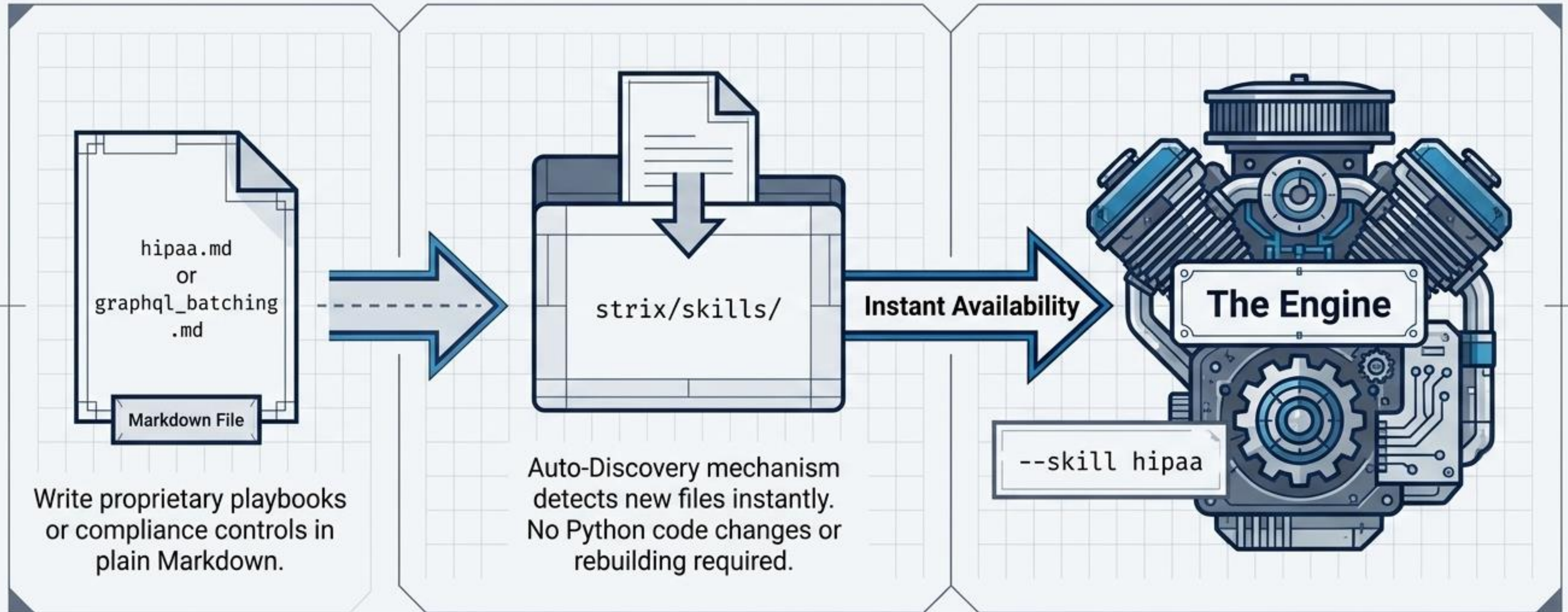


For white-box testing, source code is exposed via safe bind mounts. Agents can propose diff-style patches without compromising the host.

Guaranteeing Deterministic Coverage

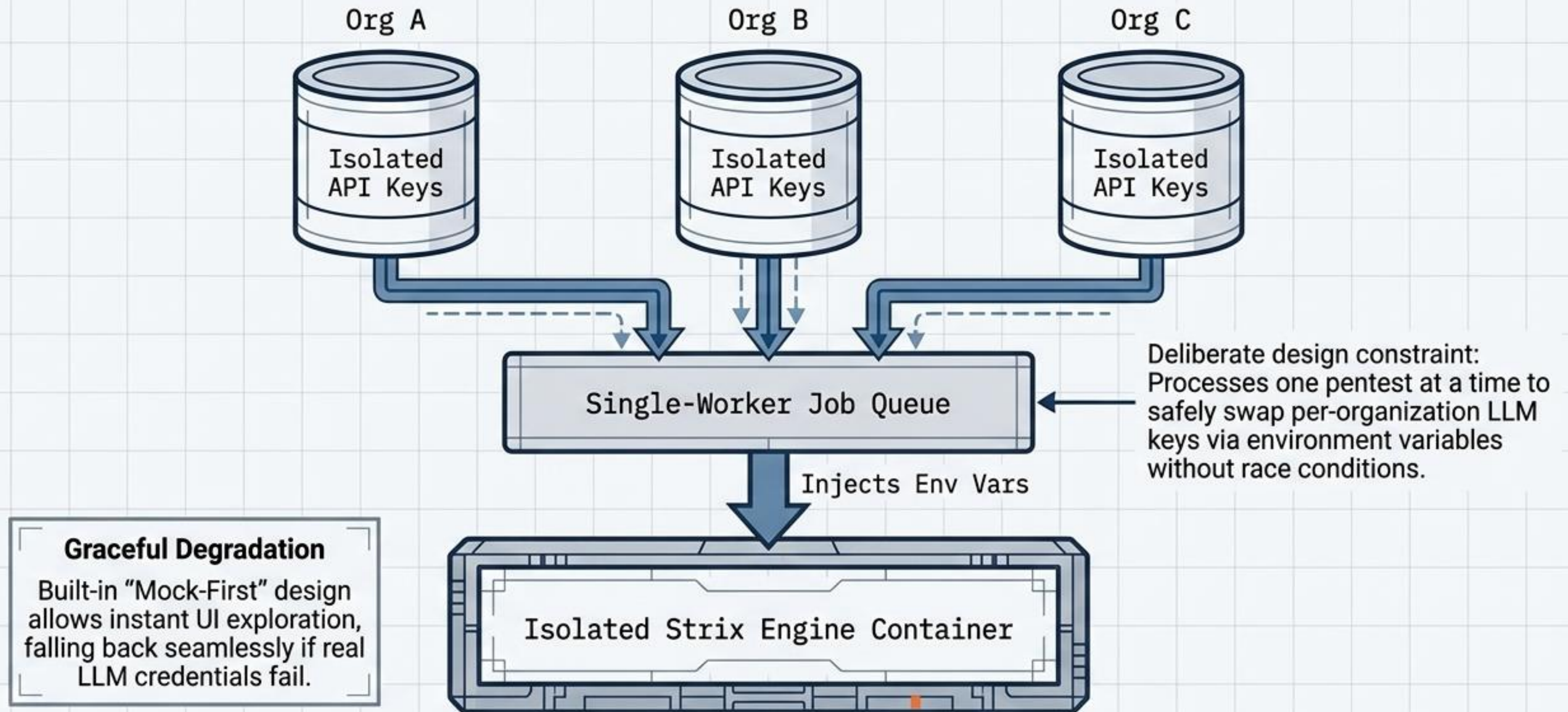


Zero-Code Extensibility

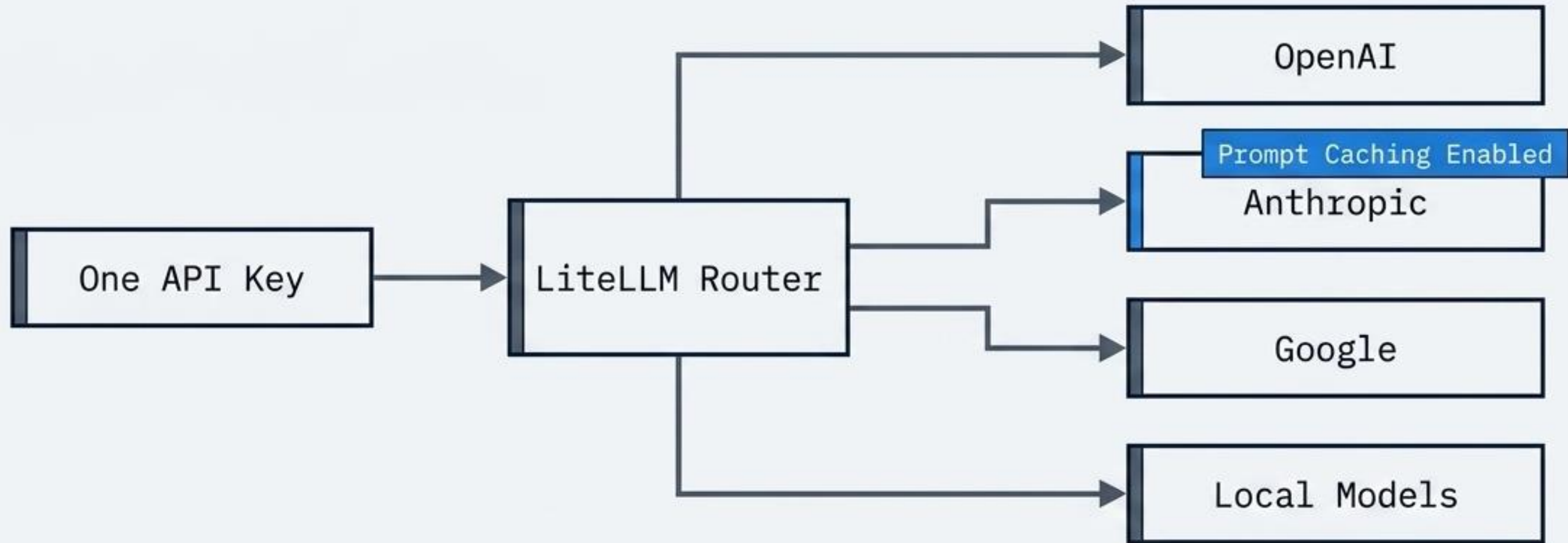


The Result: Inject zero-day tactics, internal auth playbooks, or new compliance frameworks directly into the AI's brain in seconds.

Multi-Tenancy and the SaaS Worker Queue



Agnostic Intelligence and Unit Economics



Decoupled Intelligence

Strix uses LiteLLM to remain agnostic. A single API key configuration grants access to the entire model ecosystem without locking into one provider.

Infinite Loop Protection & Caching

Custom tool-call limiters and stream-idle timeouts ensure a stalled model never drains the budget. Built-in prompt caching slashes the cost of deep context windows during extended pentests.

One Engine, Many Faces

The exact same core architecture powers radically different user experiences.

Web SaaS



`app.strix.ai`

Fully managed, multi-tenant enterprise platform for continuous security and SSO.

Local Terminal TUI



`strix --target ./`

Run natively head-less or in the terminal with highly interactive, real-time feedback.

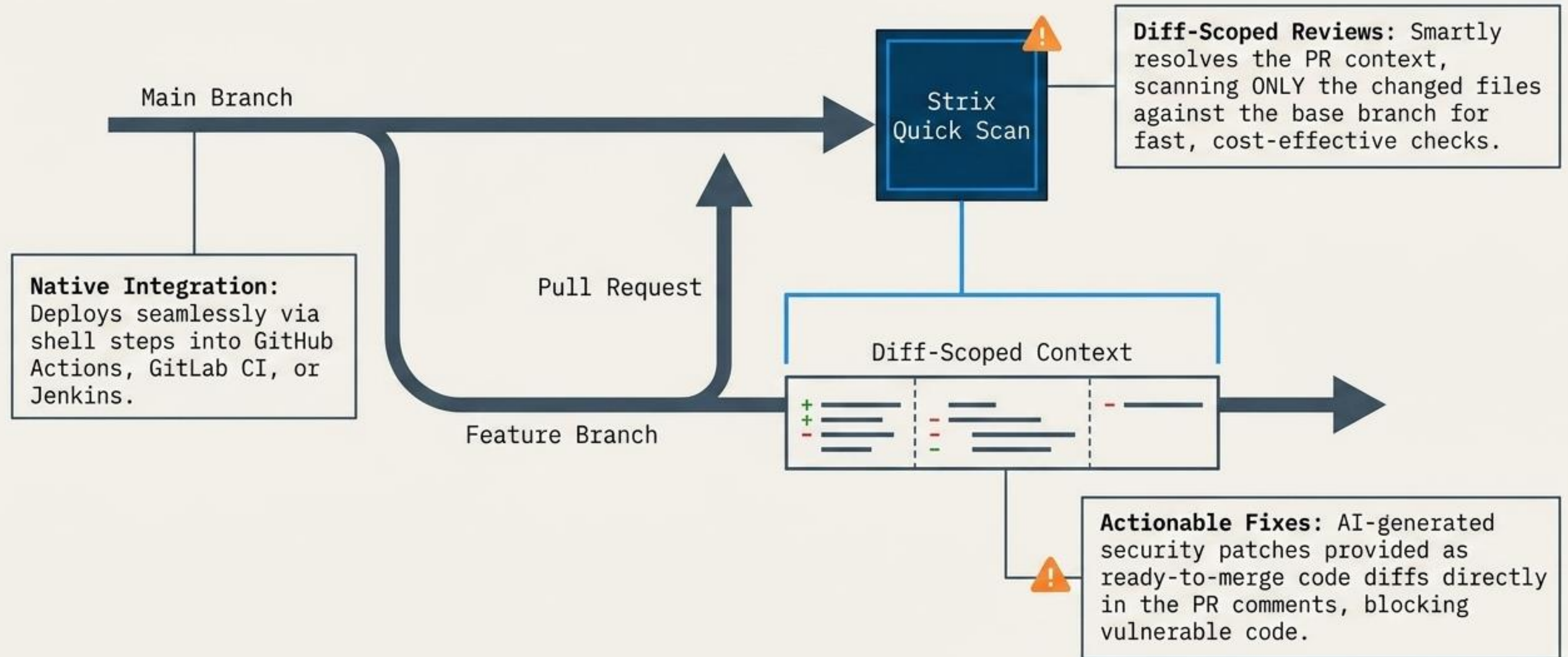
Local Web Viewer



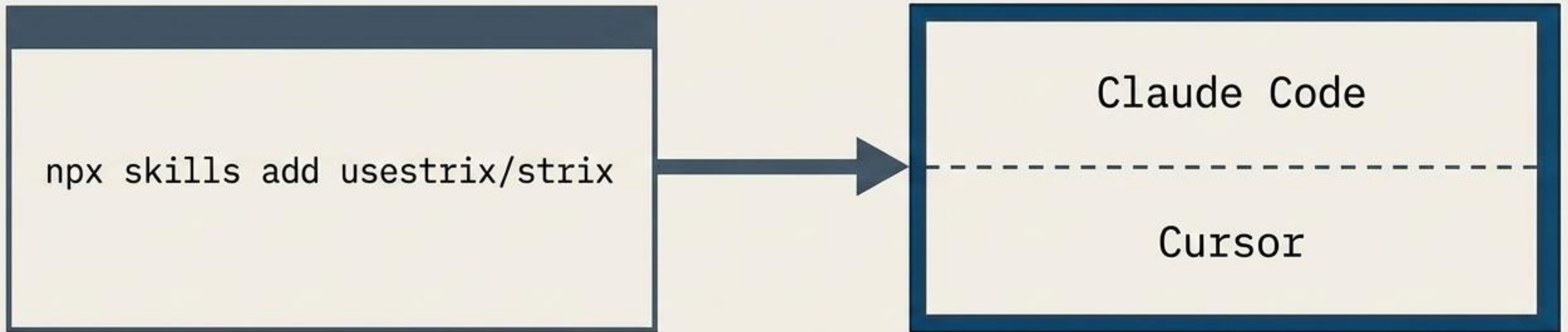
`strix view`

Absolute local privacy. Reads scan results straight off the local disk. Zero cloud upload required.

The Pipeline Guardian: CI/CD Integration



Agent-Ready Security Skills



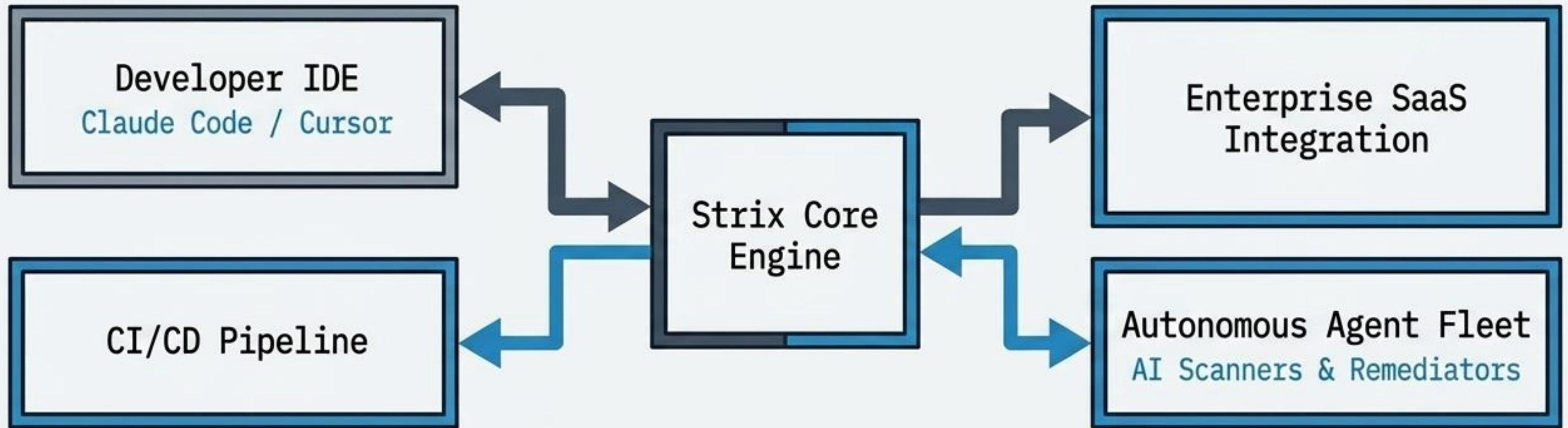
- Strix is fully compliant with the SKILL.md standard, empowering existing coding assistants to run pentests natively.
 - ▶ **Autonomous Scanning:** Agents run headless Strix scans directly within the IDE.
 - ▶ **Contextual Reading:** Agents consume the resulting vulnerability JSON straight off the local disk.
 - ▶ **Self-Healing Workflow:** Agents autonomously apply remediations to the source code and re-scan to verify the fix.

Why This Architecture Wins

	Legacy SAST/DAST	Manual Pentesting	Strix Architecture
Speed of Execution	Fast / Automated	Slow / Weeks	Fast / Minutes
False Positive Rate	High (Regex-based) ⚠️	Low (Validated)	Zero (PoC Validated) ✓
Extensibility	Custom rule writing	Requires human training	Zero-code Markdown drop-in
CI/CD Readiness	Yes (But noisy) ⚠️	No ❌	Native (Diff-scoped) ✓
Depth of Logic Testing	Static patterns only	Deep human logic	Dynamic AI logic & chained exploits

By combining the speed of automated tools, the logic of a human pentester, and an impossibly easy extensibility model, Strix provides the ultimate DevSecOps foundation.

Secure by Design. Scalable by Architecture.



Strix is not just a point-solution; it is an extensible, economically scalable infrastructure for the future of application security.

Built on a pure isolation boundary, it is ready to deploy everywhere: from the developer's IDE, to the CI/CD pipeline, to the enterprise SaaS.